EE 226: Semiconductor Devices

Lecture 2: Semiconductor Devices overview – Part 2

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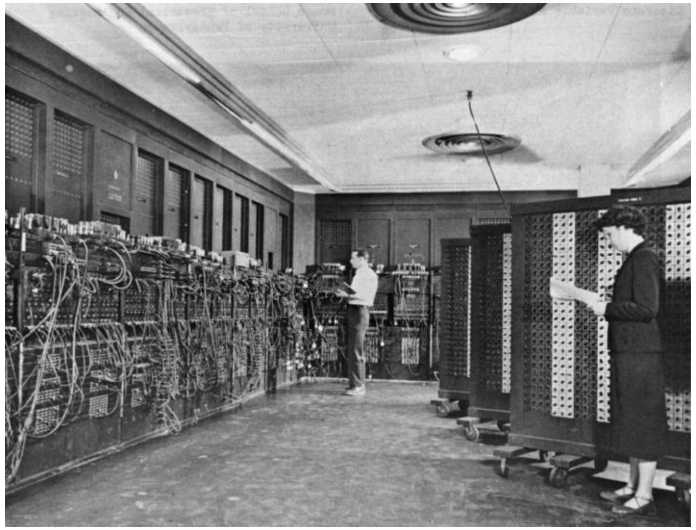
Department of Electrical Engineering
IIT Gandhinagar

Computer - Then and Now

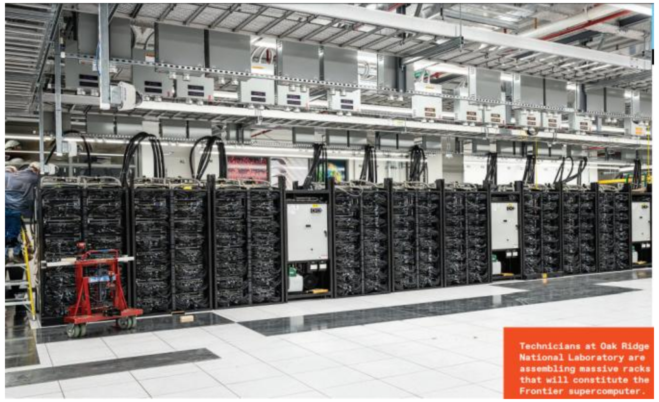
Can we make this better?

ENIAC was the first electronic, Turing-complete device



Area: 300 sq ft
Computation: 0.00289 megaflops
Power Consumption: 150 kW

Supercomputer, Oak Ridge National Laboratory



Area: ~7000 sq ft
Computation: ~1 exaflops
Power Consumption: ~20 MW

Area: ~2 sq ft
Computation: petaflops to exaflops
Power Consumption: ~10-20W

Area + Speed + Power

https://eniacday.org/

https://www.computerhistory.org/

The Power of 10!

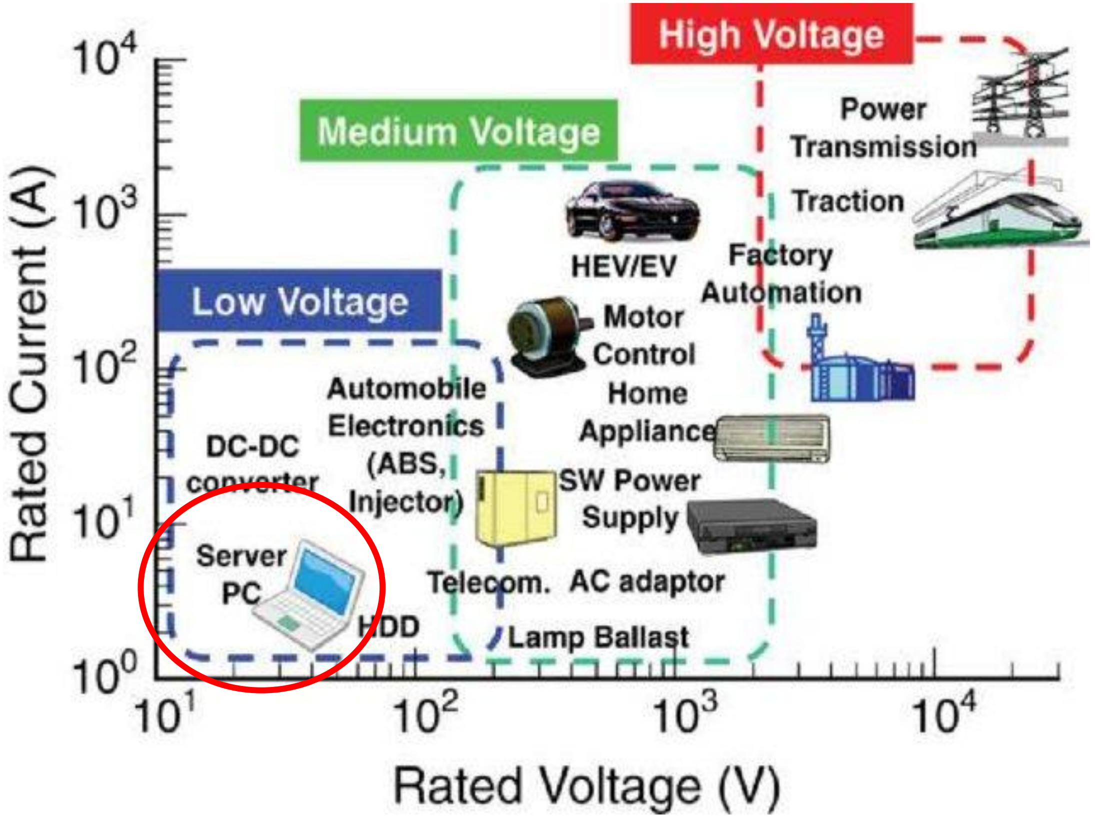
Powers of 10

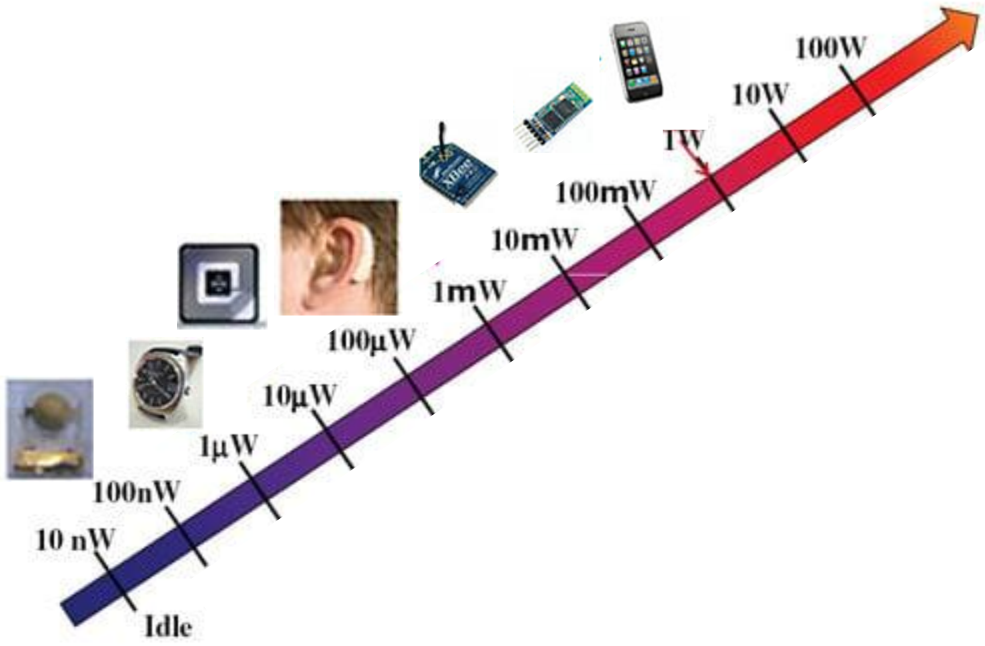
sciencenotes.org

Prefix	Exponent	Number	Scientific Notation	Name
Exa (E)	18	1,000,000,000,000,000,000	10^{18}	quintillion
Peta (P)	15	1,000,000,000,000,000	10^{15}	quadrillion
Tera (T)	12	1,000,000,000,000	10^{12}	trillion
Giga (G)	9	1,000,000,000	10^{9}	billion
Mega (M)	6	1,000,000	10^{6}	million
kilo (k)	3	1,000	10^{3}	thousand
hecto (h)	2	100	10^{2}	hundred
deca (da)	1	10	10^{1}	ten
	0	1	10^{0}	one
deci (d)	-1	0.1	10^{-1}	one tenth
centi (c)	-2	0.01	10^{-2}	one hundredth
milli (m)	-3	0.001	10^{-3}	one thousandth
micro (μ)	-6	0.000001	10^{-6}	one millionth
nano (n)	-9	0.000000001	10^{-9}	one billionth
pico (p)	-12	0.000000000001	10^{-12}	one trillionth
femto (f)	-15	0.000000000000001	10^{-15}	one quadrillionth
atto (a)	-18	0.000000000000000001	10^{-18}	one quintillionth

A quick look into various electronic devices

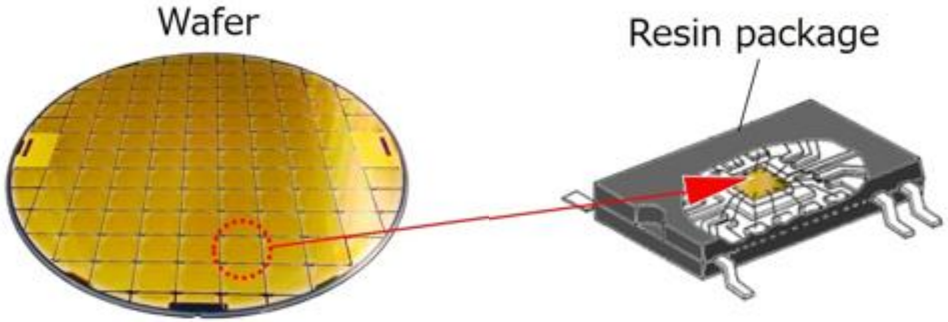
Notice the Voltage and Power ranges





M. T. Todaro *et al*in *IEEE Trans. on Nano.*, 2018,

What's inside a smartphone?

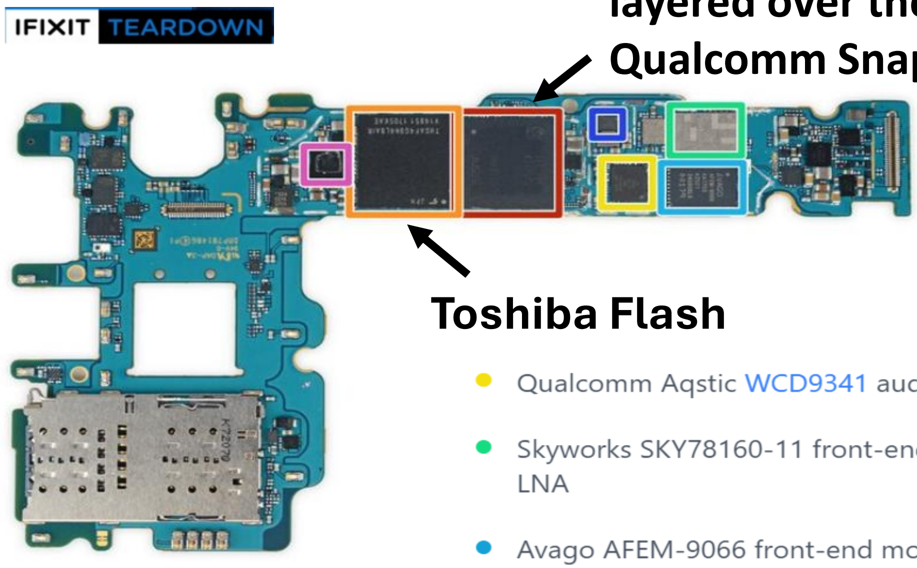


Chips are produced on wafers, and a single wafer can house numerous chips

A single chip cut from a wafer is covered in a resin package to become a semiconductor product

https://www.toshiba-clip.com/

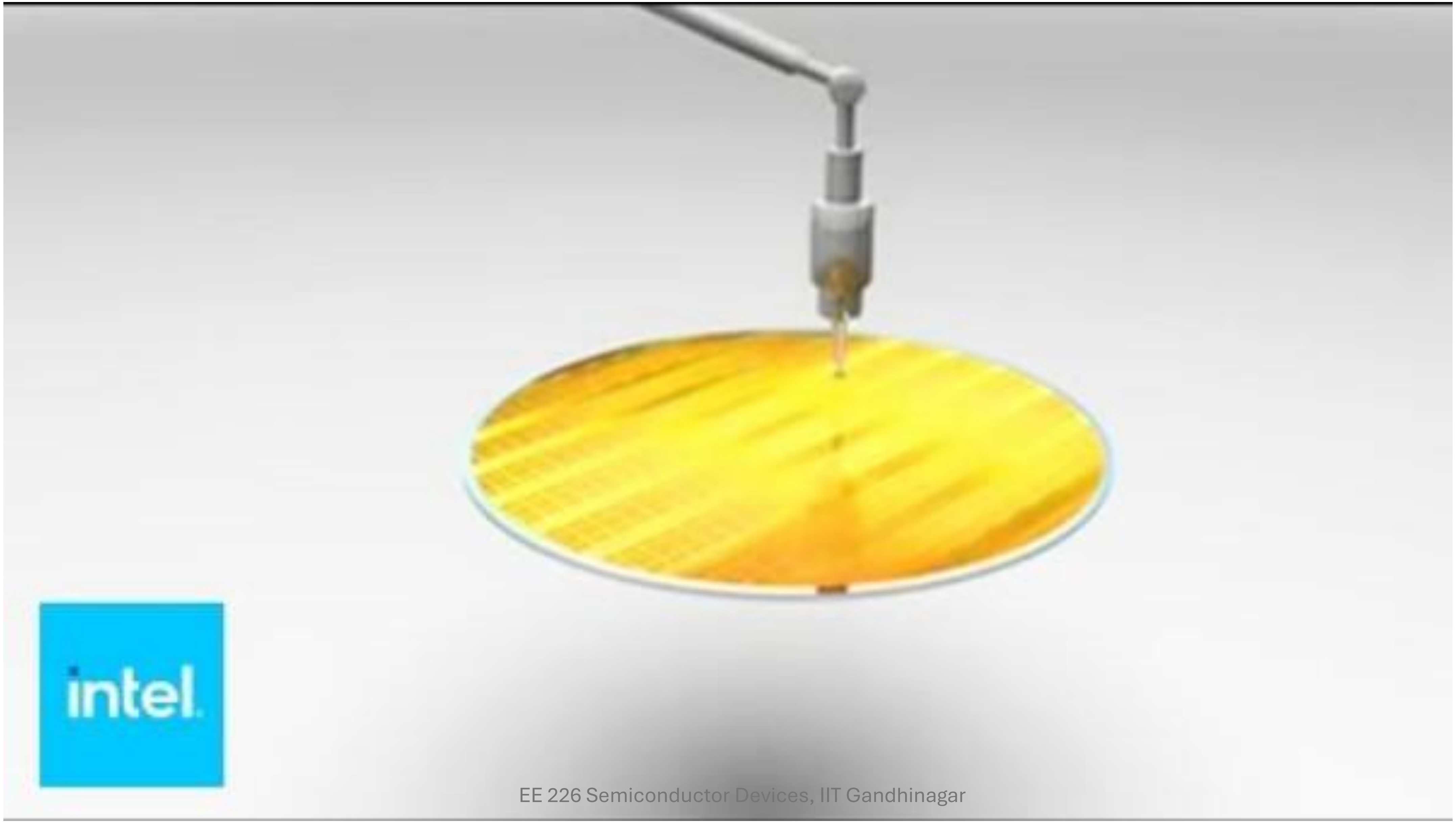
Samsung 4 GB LPDDR4 RAM layered over the Qualcomm Snapdragon



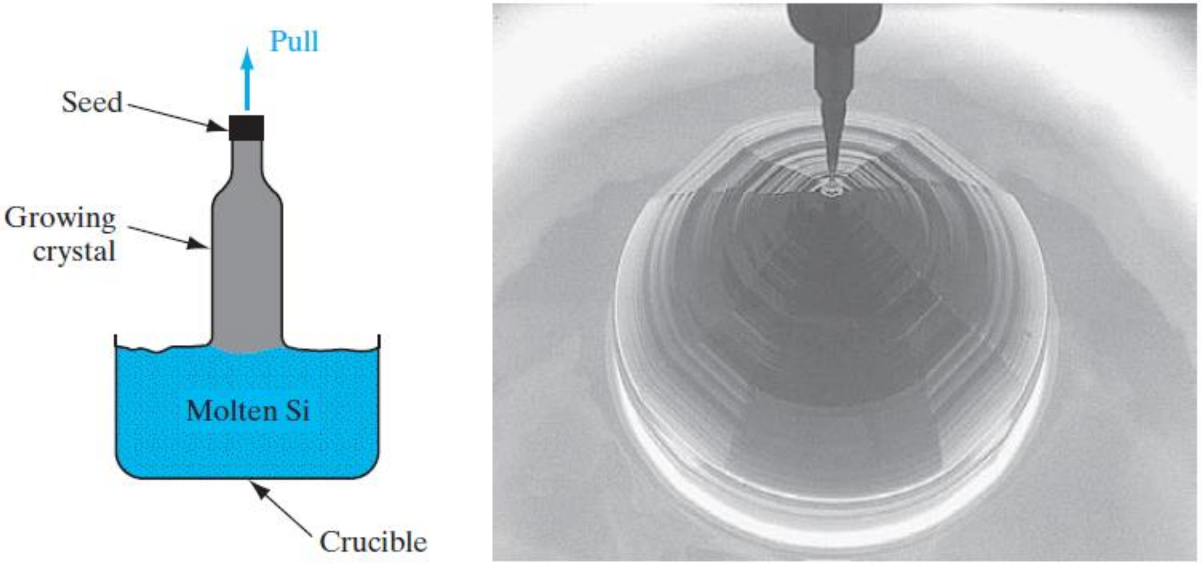
Toshiba Flash

- Qualcomm Aqstic WCD9341 audio codec
- Skyworks SKY78160-11 front-end module w/ LNA
- Avago AFEM-9066 front-end module
- Qualcomm QET4100 envelope tracker
- Silicon Mitus SM5720 Interface PMIC

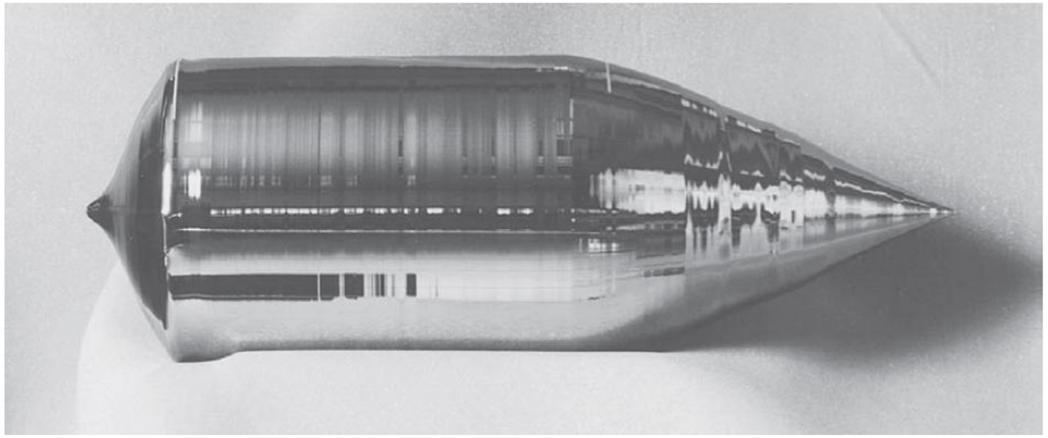
Sand to Silicon – Making of a Chip

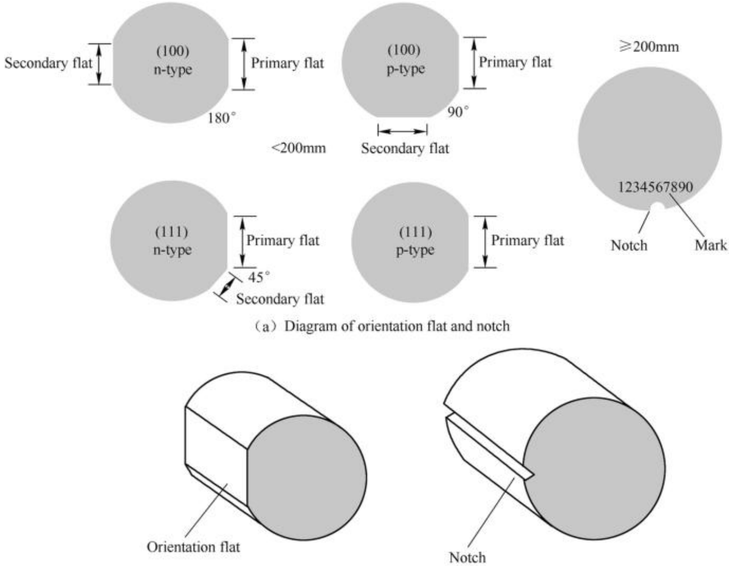


Wafer fabrication process



(a) (b)



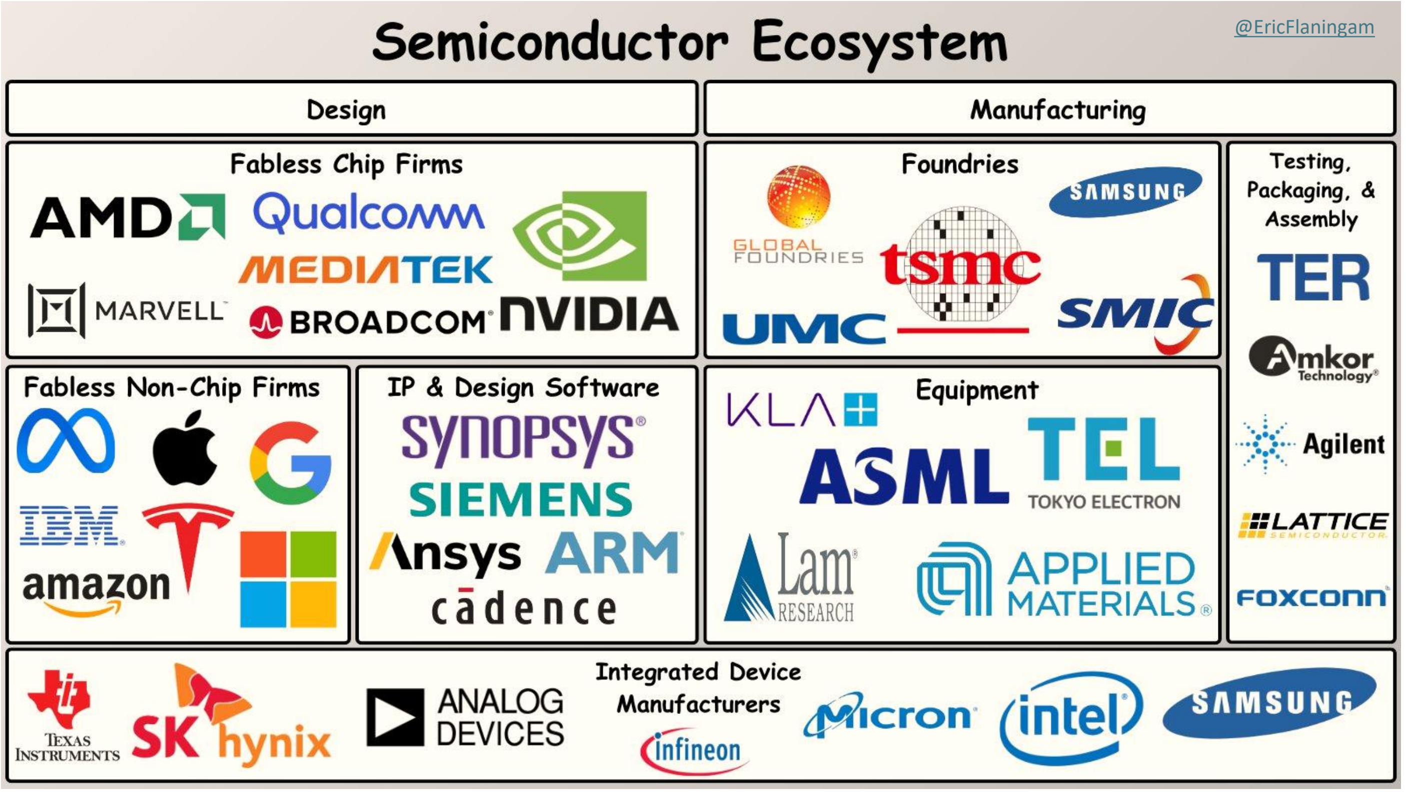


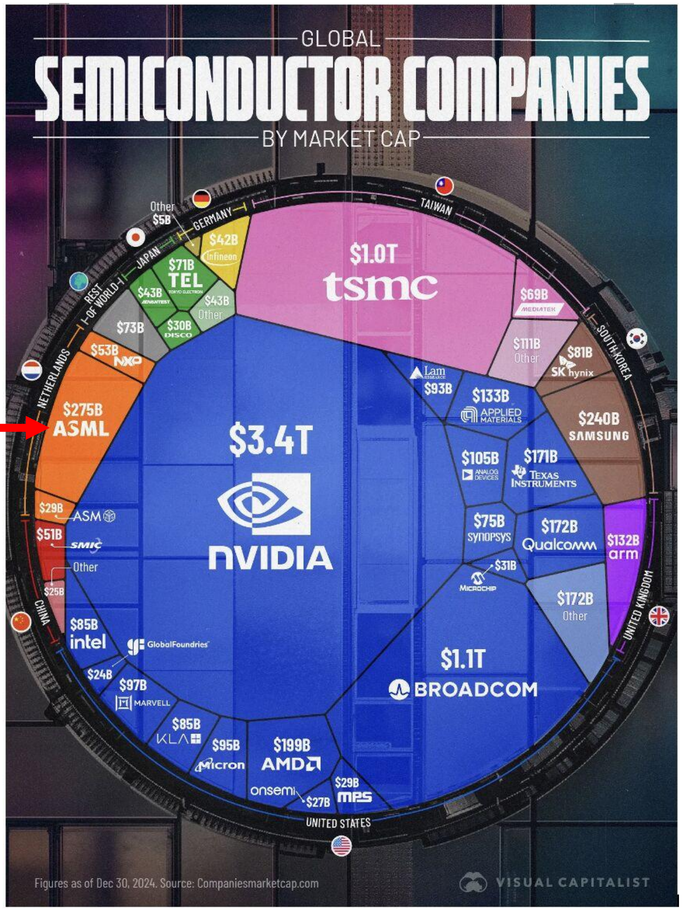
(a) Diagram of orientation flat and notch

(b) Axonometric diagram

Today’s Semiconductor Ecosystem

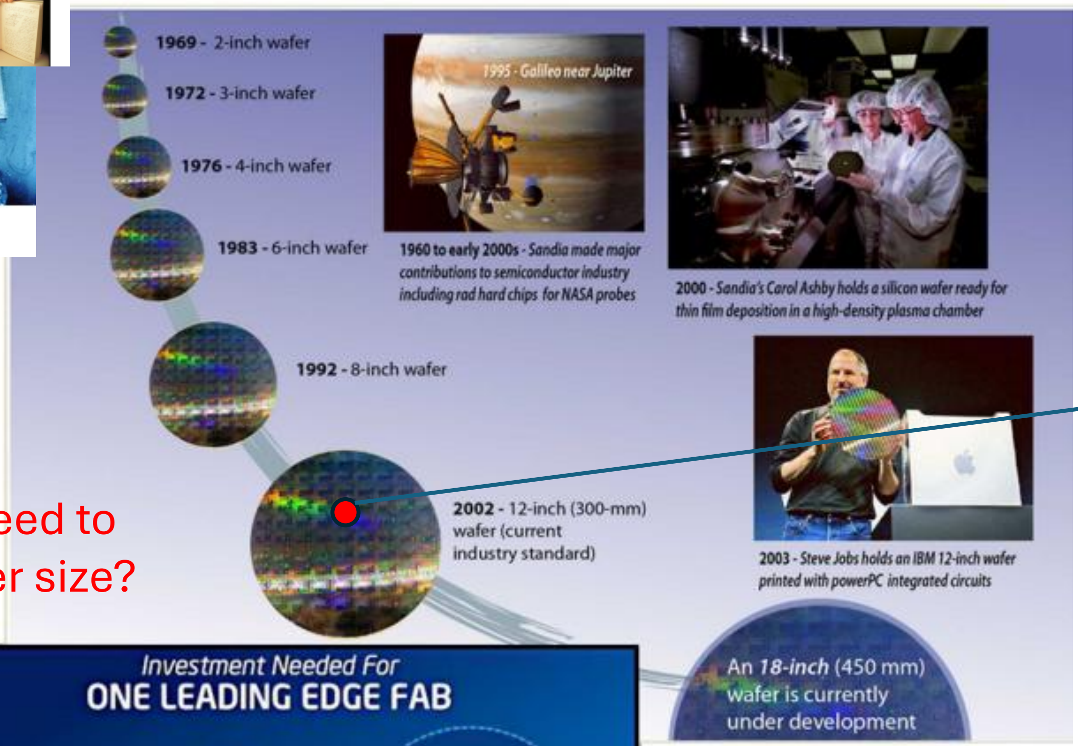
- **Fabs** (short for “fabrication”) are facilities in which semiconductor products are manufactured.
- **Fabless companies** focus on designing chips. They partner with other companies (foundries) for the manufacturing phase.
- **Foundries** are companies that manufacture semiconductor products as a service. <u>They do not design chips</u>.
- **Integrated device manufacturers (IDMs)** design and manufacture their own chips in their own fabs.





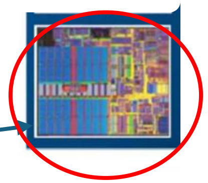
Computing – Transistors - MOSFETs

Jack Kilby
Texas Instruments

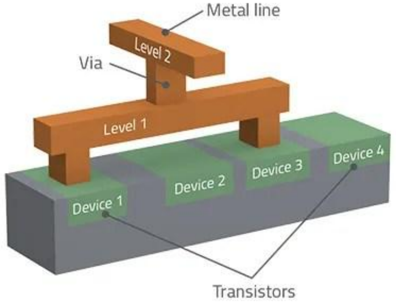


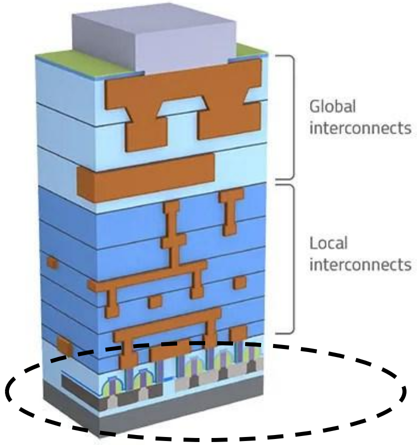
What is the need to increase wafer size?

D. Nenni, 450mm Wafers are Coming!, SemiWiki.com, Aug 14 2013



Cross sectional view

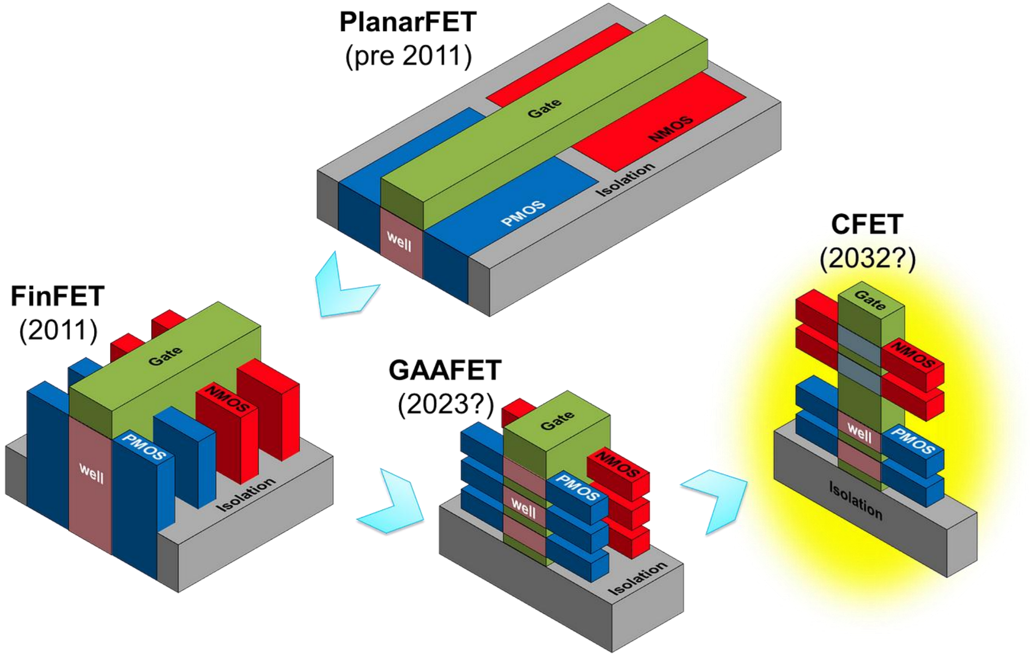


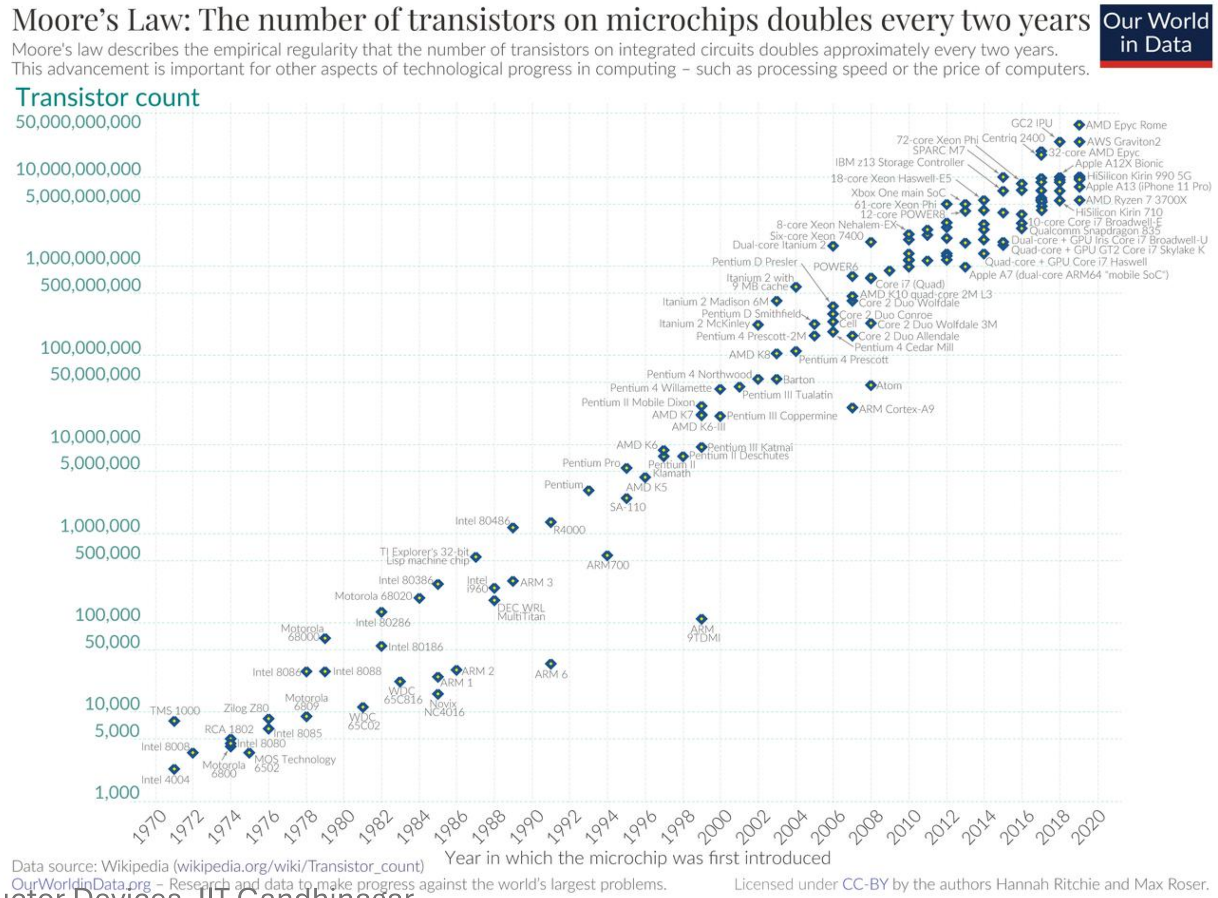


Transistors

Where are we now?

Gordon Moore, Co-founder Intel





- Towards the end of Module 4, we will understand why do we need different MOSFET innovations

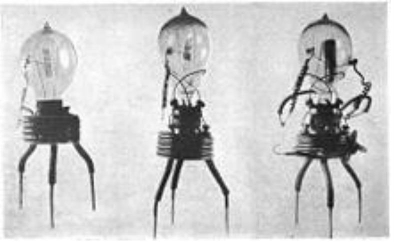
How about 100 years before...

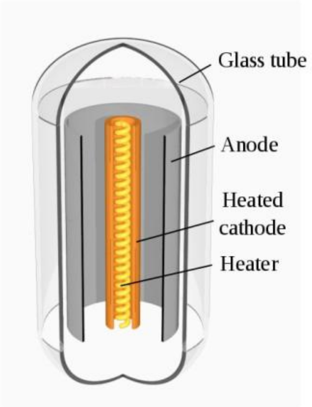
1916 to 1960

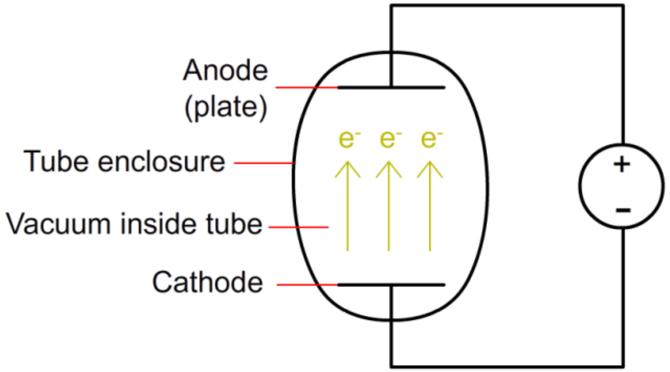
Vacuum Tubes!

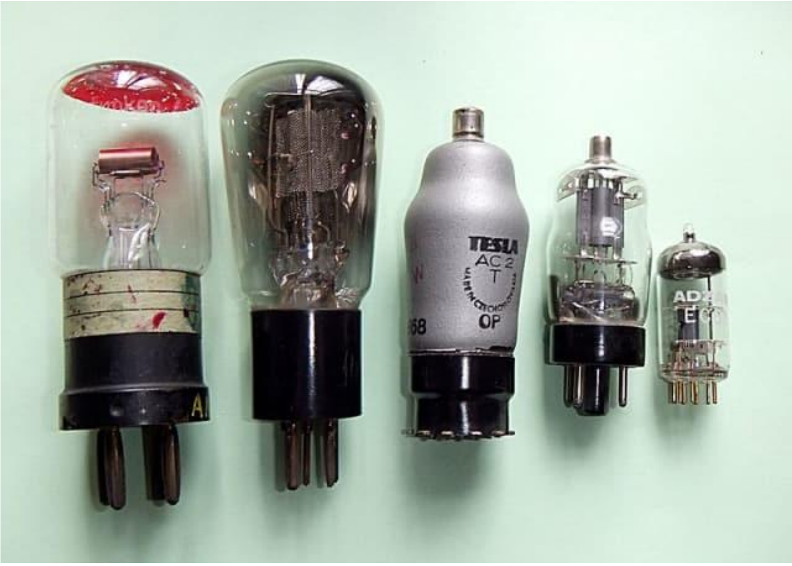
1904, Ambrose Fleming, First vacuum tube device, oscillation valve based on Thermionic Emission

Diode

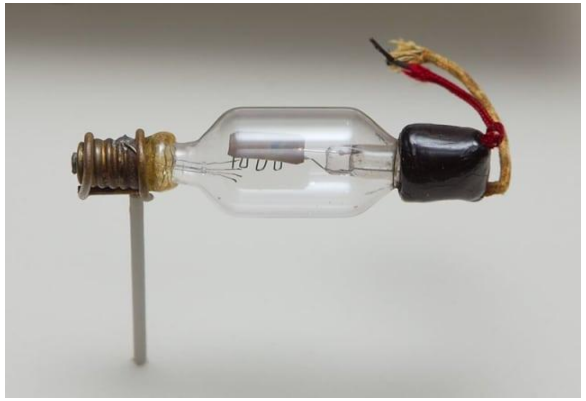


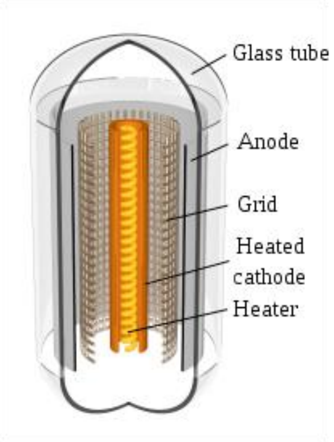


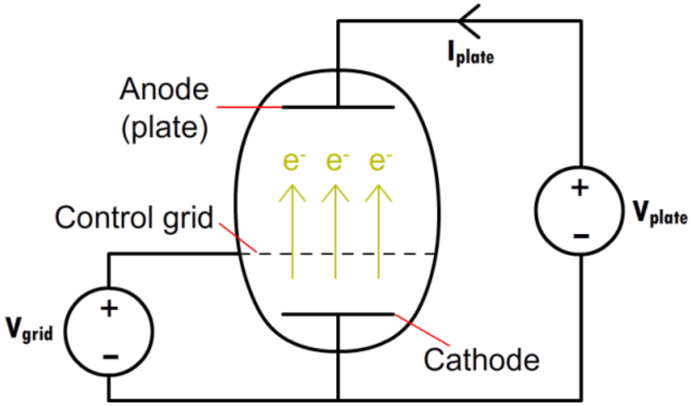




1906, The original triode vacuum tube, the Audion, invented by Lee de Forest





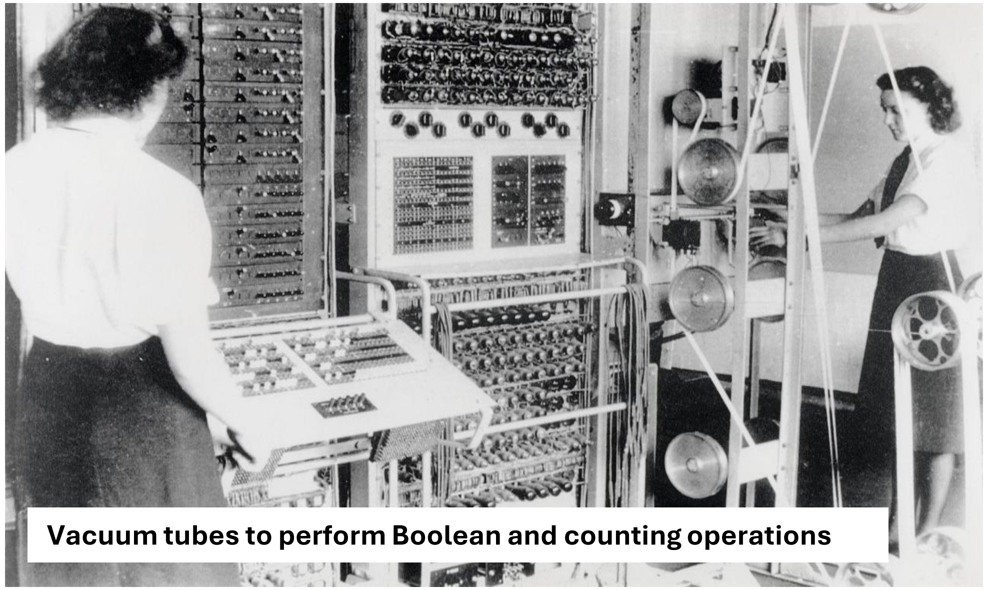


Modern use in guitar amplifiers

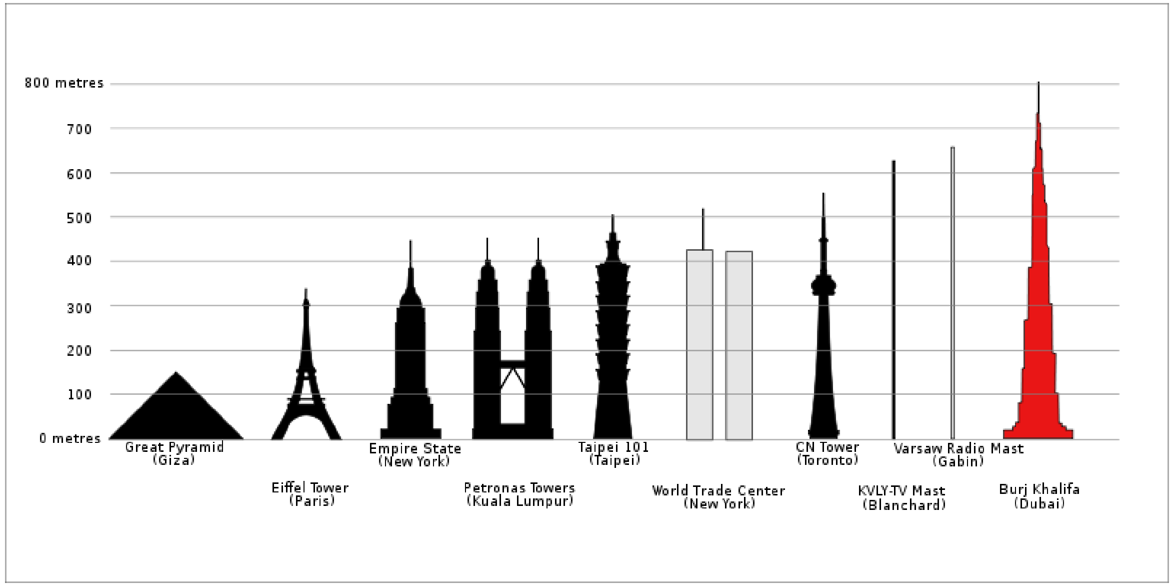
Transistor

https://www.engineering.com/story/vacuum-tubes-the-world-before-transistors

Vacuum Tubes



Colossus, the first electronic digital programmable computing device, was used to break German ciphers during World War II



22 reactors with a total capacity of ~6.78 GW

128 GB = 1Tbit

2.4cm X 3.2cm X 0.21cm

Volume : 1. 6cm³ Weight : 2g

Voltage : 2.7 - 3.6V

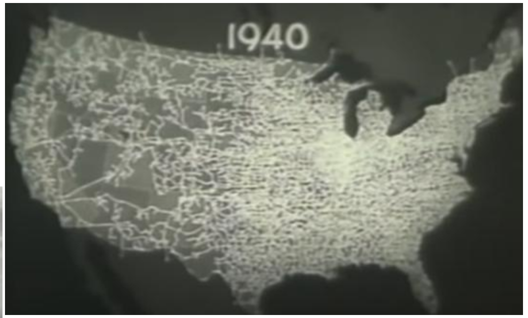
Old Vacuum Tube : 1-bit

5cm X 5cm X 10cm, 100g, 50W

What are volume, weight, power consumption for 1Tbit

Invention of First Transistor!

Communication Network

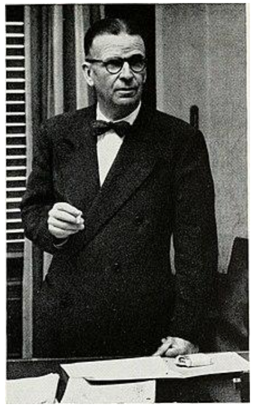


World war 2!

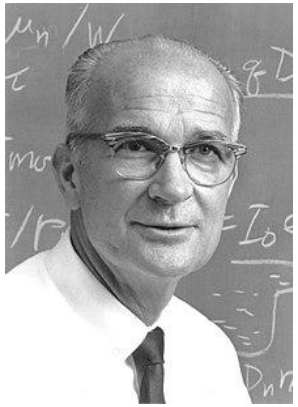
Radar's

Needed a device to convert radar signal to something which can be measured!

Mervin Kelly, Bell Labs
Director of research, president, and chairman of the board of directors Bell Labs



William Shockley
Theoretical Physicist
PhD from MIT in 1936



Walter Brattain
Experimental Physicist
Ph.D. University of Minnesota 1929

John Bardeen
Experimental Physicist
Ph.D. in physics from Princeton University

And more chemist, physicist and engineers

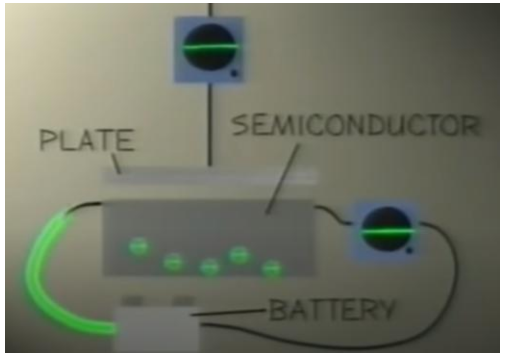
1956 Shockley, Brattain, and Bardeen won the Noble Prize for physics

https://www.youtube.com/watch?v=U4XknGqr3B

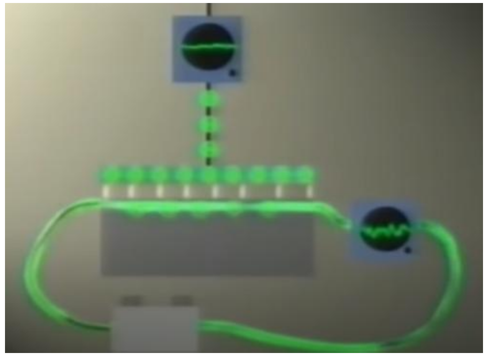
Invention of First Transistor!

The theory – Field Effect

OFF

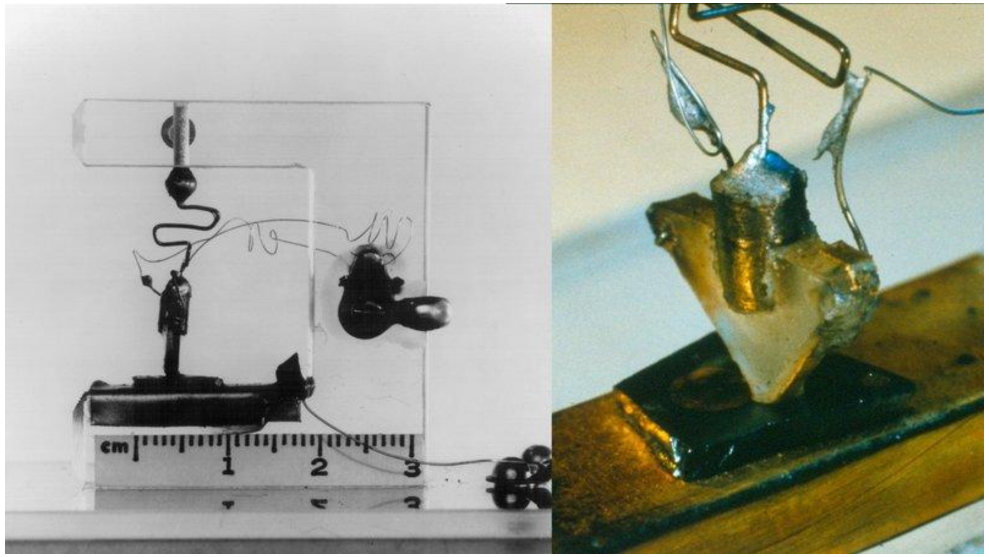


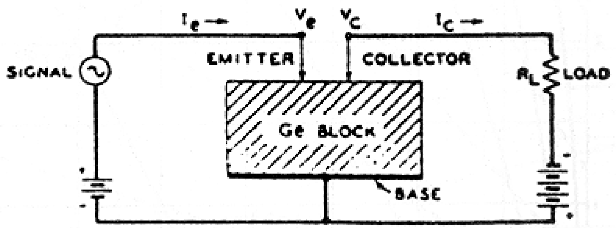
ON



Controlled ON and OFF

Experiment to Invention





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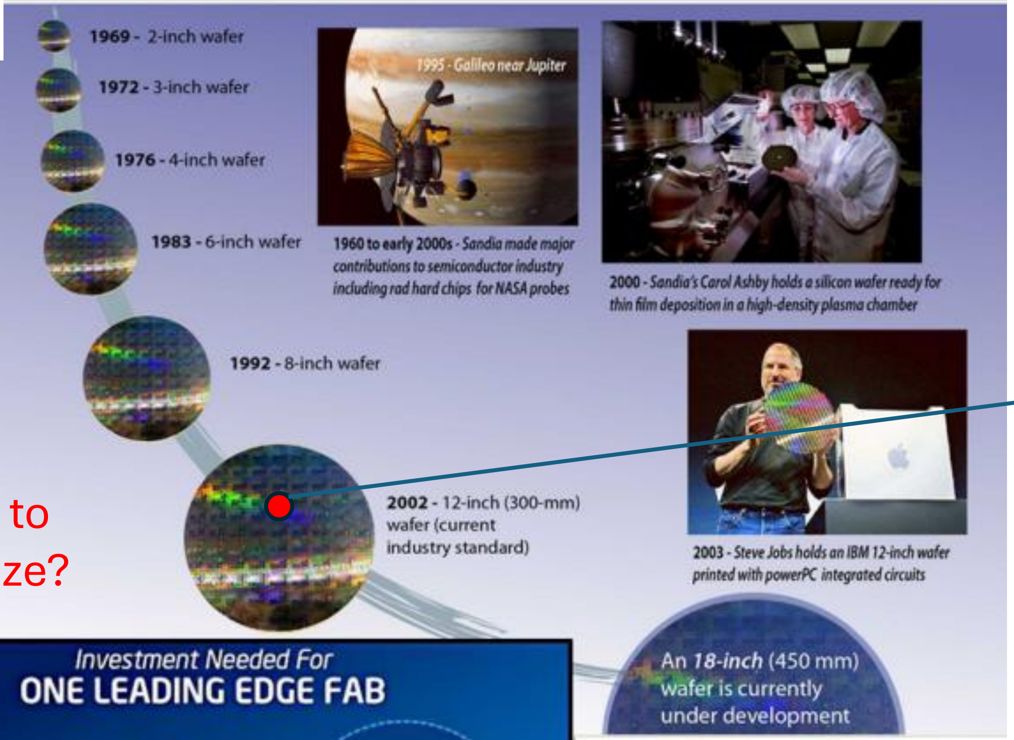

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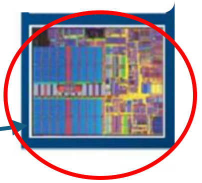

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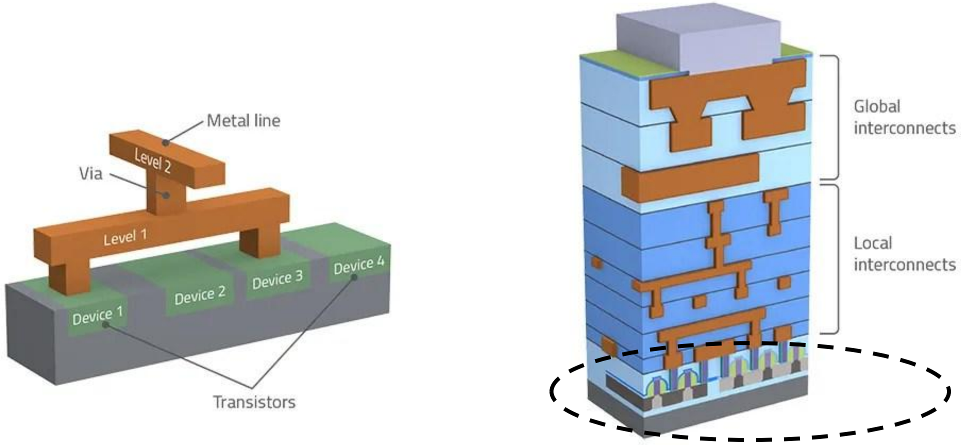

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sciencenotes.org

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Switch – How does the I-V look in linear and log scales